Q) What are the advantages of using functions?

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A) 1) Using functions it becomes easier to write programs and keep track of what they are doing. If the operation of a program can be divided into separate activities, and each activity placed in a different function, then each could be written & checked more or less independently.

2) Separating the code into modular functions makes the program easier to design and understand.

3) The length of a source program can be reduced by using functions at appropriate places.

4) It is easy to locate and isolate a faulty function for further investigations.

5) Writing functions avoids rewriting the same code over and over.

Q) What do you mean by call by value and call by reference? Explain with example.

Ans) There can be 2 types of function calls —

i) Call by value

ii) Call by reference

Arguments can be passed to functions in one of the following 2 ways:-

1) Sending the values of the arguments :-

This method is known as "Call by value". In this method, the 'value' of each of the actual arguments in the calling function is copied into corresponding

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formal arguments of the called function. In this method, the changes made to the formal arguments in the called function have no effect on the values of actual arguments in the calling function.

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2) Sending the addresses of the arguments :-

This method is known as "Call by Reference". In this method, the addresses of actual arguments in the calling function are copied into formal arguments of the called function. This means that using these addresses we would have access to the actual arguments and hence we would be able to manipulate them. So, in this method, the changes made to the formal arguments in the called function will be reflected on the values of actual arguments in the calling function.

Example of Call by Value :-

```
void main()
{
    int a=10, b=20;
    swapv(a, b);
    printf("\n a=%d, b=%d", a, b);
    getch();
}
swapv(int x, int y)
{
    int temp;
    temp = x;
    x = y;
    y = temp;
    printf("\n x=%d, y=%d", x, y);
}
```

Output :-

x=20, y=10

a=10, b=20

Example of Call by Reference

```
void main()
{
    int a=10, b=20;
    swapr(&a, &b);
    printf("\n a=%d, b=%d", a, b);
    getch();
}
swapr(int *x, int *y)
{
    int temp;
    temp = *x;
    *x = *y;
    *y = temp;
    printf("\n %d, %d", *x, *y);
}
```

Output :-

20, 10

a=20, b=10

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Here, we can see that in case of call by value, the values of a and b remain unchanged even after exchanging the values of x and y.
But, in case of call by reference, ~~the values of a and b have been swapped~~ it has been possible to swap the values of a and b using their addresses stored in x and y.

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